

LAGGING WELLS

LAG TIME STUDY BETWEEN CYGNET AND IVM
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INTRODUCTION:

The intention of this document is to point out the findings from a study done to determine how long it takes (aka “Lag-Time”) for data to transmit from Cygnet to IVM. Normally, IVM gets its data from Proficity, but Proficity was eliminated to isolate the time it takes for the data to transmit directly from Cygnet.

SUMMARY OF PROCESS EMPLOYED TO DETERMINE “LAG-TIME”

1. CREATE SAMPLE WELL LIST
2. CREATE "OFFENDER WELLS" LIST
3. DEMAND SCAN "OFFENDER WELLS" IN IVM AND RECORD RETURNED DATA
4. POLL "OFFENDER WELLS" IN CYGNET AND RECORD RETURNED DATA
5. COMPARE BOTH SETS OF RETURNED DATA

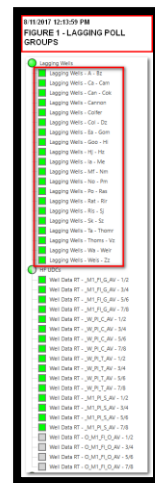
DETAILS OF PROCESS EMPLOYED TO DETERMINE “LAG-TIME”

SAMPLE WELL LIST

IVM currently collects data for 1,350 wells. A majority, of the wells within IVM, receives data from Cygnet in an expeditious manner. To ensure maximum efficiency, the IAM team is required to use alternate techniques for a number of wells that do not come back as quickly. Currently, 263 of the 1'350 wells are "housed" within a specially created "Group" called "Lagging Wells".

DETAILS ABOUT THE “SAMPLE WELLS” aka “LAGGING WELLS”:

- The 263 wells experience a “Lag-Time” of at least 90 minutes.
- Currently, we are required to maintain multiple “Groups”
 - “LAGGINGS WELLS” and “HF UDCs”.
 - Please see FIGURE 1 to the right.
- The “Groups” are then further separated into “Poll Groups”, which allow for polling that is more efficient.
 - The wells in the “Lagging Wells” are separated by alphabetic order.
 - The wells in the “HF UDCs” are separated by the site service the well belongs to.
 - This can be seen with a red border around them in FIGURE 1.
 - The poll groups are limited to a predetermined number per poll group.
- To further maximize efficiencies, the IAM team would like to assimilate the aforementioned “Lagging” Poll Groups into the “HF UDCs” Poll Groups (also seen in FIGURE 1), but until the “Lag-Time” is reduced to ensure maximum efficiency, they will need to be separated.
- The list of these 263 wells is found within the Appendix of this document called Exhibit 1.



CREATE “OFFENDER WELLS” LIST

To expedite the study, the list was narrowed down to six wells called the “OFFENDER WELLS”; the list of “OFFENDER WELLS” is found within the Appendix of this document called Exhibit 2.

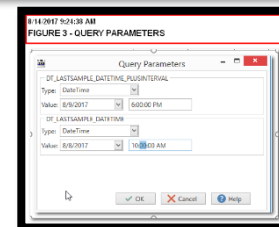
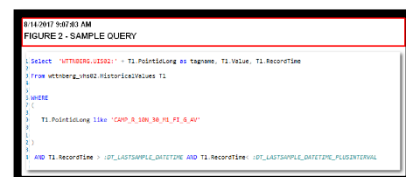
DETAILS ABOUT THE “OFFENDER WELLS”:

- These wells are known to have the greatest amount of “Lag-Time”.
- They are from multiple routes.
- They come from different horizontal pads.
- They belong to from multiple site services.

DEMAND SCAN “OFFENDER WELLS” IN IVM AND RECORD RETURNED DATA:

Demand Scanning is the process employed by IVM to query data from the data source the user set-up up within the query. A sample of the query can be seen in Figure 2.

- Currently, IVM polls Proficy directly. To remove any possibilities that Proficy was the cause of the delay, the “Demand Scan” was set to poll Cygnet directly.
- The site service is part of the query. (See Figure 2)
- One tag is used to query Cygnet. (See Figure 2)
- Query Parameters are set for the time and date. (See Figure 3.)



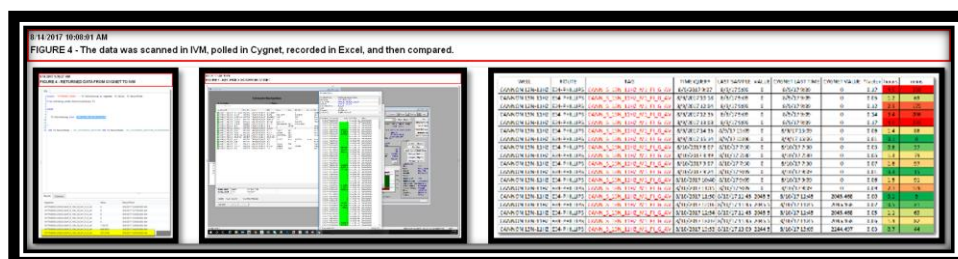
POLL “OFFENDER WELLS” IN CYGNET AND RECORD RETURNED DATA:

Using Cygnet Vision, wells were polled and the results were recorded in Excel.

- In order to eliminate the possibility of the time of day the tag was polled as the cause, the tag was polled simultaneously in Cygnet and IVM.

COMPARE BOTH SETS OF RETURNED DATA

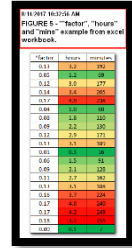
The data was scanned in IVM, polled in Cygnet, recorded in Excel, and then compared.



- The results were compared by using the time of the scan (current time/date) in both Cygnet and IVM.

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- This “current date/time” was subtracted from the “last date/time” stamp of each.
 - This value represents the “Lag-Time” data was received from Cygnet in both IVM and Cygnet.
 - To better represent, the value was then multiplied (“*factor” column in the Excel workbook) by 24 to calculate hours and then by 60 to calculate minutes. (See Figure 5)
- The well name, route, and a value for both IVM and Cygnet were recorded to show that the data being compared was in fact the same.



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FIGURE 6 - Well name, route, and values in excel workbook.

WELL	ROUTE	TAG	TIME QUERY	LAST SAMPLE	VALUE	CYGNET LAST TIME	VALUE
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/9/2017 8:57	8/9/17 5:45	357.5149	8/9/17 5:45	357.51

RESULTS

Multiple “OFFENDER WELLS” had a “Lag-Time” of as much as 255 minutes; average “Lag-Time” was 130 minutes.

The following table shows the results from the “scanning” and “polling” for one of the six wells.

WELL	ROUTE	TAG	TIME QUERY	LAST SAMPLE	VALUE	CYGNET LAST TIME	VALUE	*factor	hours	minutes
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/9/2017 8:57	8/9/17 5:45	357.5149	8/9/17 5:45	357.51	0.13	3.2	192
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/9/2017 10:15	8/9/17 9:06	0	8/9/2017 9:06	0.00	0.05	1.2	69
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/9/2017 12:03	8/9/17 9:06	0	8/9/2017 9:06	0.00	0.12	3.0	177
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/9/2017 12:31	8/9/17 9:06	0	8/9/2017 9:06	0.00	0.14	3.4	205
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/9/2017 13:04	8/9/17 9:06	0	8/9/17 9:06	0	0.17	4.0	238
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/9/2017 14:21	8/9/17 13:21	0	8/9/17 13:21	0	0.04	1.0	60
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/9/2017 15:11	8/9/17 13:21	0	8/9/17 13:21	0	0.08	1.8	110
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 8:04	8/10/17 5:54	958.448	8/10/17 5:54	958.448	0.09	2.2	130
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 8:45	8/10/17 5:54	958.448	8/10/17 5:54	958.448	0.12	2.9	171
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 9:03	8/10/17 5:54	958.448	8/10/17 5:54	958.448	0.13	3.1	189
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 9:22	8/10/17 9:06	0	8/10/17 9:06	0	0.01	0.3	16
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 10:37	8/10/17 9:06	0	8/10/17 9:06	0	0.06	1.5	91
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 11:14	8/10/17 9:06	0	8/10/17 9:06	0	0.09	2.1	128
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 11:48	8/10/17 9:06	0	8/10/17 9:06	0	0.11	2.7	162
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 12:14	8/10/17 9:06	0	8/10/17 9:06	0	0.13	3.1	188
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 12:50	8/10/17 9:06	0	8/10/17 9:06	0	0.16	3.7	224
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 13:06	8/10/17 9:06	0	8/10/17 9:06	0	0.17	4.0	240
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 13:15	8/10/17 9:06	0	8/10/17 9:06	0	0.17	4.2	249
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 13:21	8/10/17 9:06	0	8/10/17 9:06	0	0.18	4.3	255
CAMP 10N-30HZ	NHS-SAATHOFF	CAMP_R_10N_30_M1_FI_G_AV	8/10/2017 13:31	8/10/17 13:24	0	8/10/17 13:24	0	0.00	0.1	7

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APPENDIX

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SAMPLE WELLS - WELLS "HOUSED" IN THE "LAGGING WELLS" POLL GROUP

ObjectInstanceName										
Abbey State V16-77HN	Brown 37C-26HZ	Cannon 13C-10HZ	Dove 22N-13HZ	Gobbler 37N-W22HZ	Jelts 36N-19HZ	Nichols 3C-5HZ	Powers 3C-27HZ	Russian Fiddler 13N-18HZ	Thomason 2C-9HZ	Weichel 26N-15HZ
Aspen State V16-75HN	Brown Federal 26N-35HZ	Cannon 13N-10HZ	Dove 24N-13HZ	Gobbler 3C-23HZ	Jelts 37C-19HZ	Nichols 3N-5HZ	Powers 3N-27HZ	Russian Fiddler 14N-18HZ	Thomason 2N-9HZ	Weichel 27C-11HZ
Barclay 15N-28HZ	Camp 10N-30HZ	Cannon 13N-11HZ	Dove 25C-13HZ	Gobbler 3N-23HZ	Jelts 37N-19HZ	Nichols 4N-5HZ	Powers 4C-27HZ	Russian Fiddler 15N-18HZ	Thomason 34N-9HZ	Weichel 27N-11HZ
Barclay 16N-28HZ	Camp 15N-26HZ	Cannon 14C-10HZ	Dove 25N-13HZ	Griffiths 14N-20HZ	Jelts 39C-19HZ	Niles 28N-28HZ	Powers 4N-27HZ	Russian Fiddler 35C-18HZ	Thomason 35C-15HZX	Weichel 28N-11HZ
Barclay 1N-27HZ	Camp 1N-25HZX	Cannon 14N-10HZ	Dove 32N-13HZ	Griffiths 34N-20HZ	Marrs 28C-28HZ	Niles 29C-28HZ	Ralph 14N-19HZ	Russian Fiddler 35N-18HZ	Thomason 35N-15HZ	Weichel 2N-11HZ
Barclay 27N-27HZ	Camp 24C-30HZ	Cannon 15N-3HZ	Dove 40N-13HZ	Griffiths 35C-20HZ	Marrs 2N-28HZ	Niles 29N-28HZ	Ralph 34C-19HZ	Russian Fiddler 36C-18HZ	Van Portfliet 14N-E3HZ	Weichel 30N-23HZ
Barclay 28C-27HZ	Camp 24N-30HZ	Cannon 16C-3HZ	Dove 5C-13HZ	Griffiths 35N-20HZ	Marrs 3N-28HZ	Niles 3N-28HZ	Ralph 35C-19HZ	Russian Fiddler 36N-18HZX	Van Portfliet 35C-W3HZ	Weichel 34C-11HZ
Barclay 28N-27HZ	Camp 25N-30HZ	Cannon 16N-3HZ	Dove 7N-13HZ	Griffiths 36C-20HZ	McGuckin X05-69HN	Niles 4N-28HZ	Ralph 35N-19HZ	Russian Fiddler 37C-18HZ	Van Portfliet 36N-3HZ	Weichel 35N-11HZR
Barclay 29C-27HZ	Camp 27C-25HZ	Cannon 1N-11HZ	Dove 8C-13HZ	Haymaker 28N-9HZ	Millensifer Fed 36N-30HZ	Norton 13N-29HZ	Ralph 36C-19HZ	Russian Fiddler 37N-18HZ	Van Portfliet 38N-10HZ	Weiss 29C-33HZ
Barclay 29N-27HZ	Camp 28N-25HZ	Cannon 2C-11HZ	Encore 26N-12HZ	Haymaker 29N-9HZ	Millensifer Fed 37N-E30HZ	Norton 14N-29HZ	Rattler 4C-34HZ	Sarchet 15N-32HZ	Varra P31-74HN	Weiss 29N-33HZ
Barclay 2N-27HZ	Camp 28N-E25HZ	Cannon 34C-10HZ	Encore 40N-12HZ	Haymaker 3N-9HZ	Millensifer Fed 37N-W30HZ	Norton 34N-29HZ	Raymond 15C-29HZ	Sarchet 16N-32HZ	Varra P31-76HN	Weiss 31N-33HZ
Barclay 36N-28HZ	Camp 29N-25HZ	Cannon 35C-10HZ	Encore 41N-12HZ	Haymaker 4C-9HZ	Miracle 13C-12HZ	Norton 35C-29HZ	Raymond 15N-29HZ	Sarchet 36N-32HZ	Wardell 26N-29HZ	Weiss 3N-33HZ
Barclay 37N-28HZ	Camp 29N-31HZ	Cannon 35N-10HZ	Encore 8C-12HZ	Haymaker 4N-9HZ	Miracle 33N-12HZ	Norton 35N-29HZ	Raymond 16N-29HZ	Sarchet 36N-32HZX	Wardell 27C-29HZ	Weiss 4N-33HZ
Barclay 38N-28HZ	Camp 2C-25HZ	Cannon 36N-11HZ	Folley North #1	Howard 10C-29HZ	Miracle 40C-11HZX	Palayo 13C-11HZ	Raymond 16N-30HZ	Sarchet 37C-32HZ	Wardell 27N-29HZ	
Barclay 3N-27HZ	Camp 36N-26HZ	Cannon 36N-E3HZ	Folley South #1	Howard 24N-29HZ	Miracle 40N-11HZ	Palayo 13N-11HZ	Raymond 36N-29HZ	Sarchet 37N-32HZ	Wardell 28N-20HZ	
Barclay 4C-27HZ	Camp 37N-26HZ	Cook 36N-3HZ	Folley South #3	Howard 9C-29HZ	Miracle 9C-11HZ	Palayo 14N-11HZ	Raymond 37N-29HZ	Shurview 13N-34Z	Wardell 29N-29HZ	
Barclay 4N-27HZ	Camp 38C-26HZ	Cook 37C-3HZ	Gobbler 15N-22HZ	Hudson 13C-35HZ	Nichols 1N-5HZ	Palayo 35N-E11HZ	Raymond 38C-29HZ	Shurview 33C-3HZ	Wardell 2G-19HZ	
Brown 13C-26HZ	Camp 38N-26HZR2	Cook 37N-E3HZ	Gobbler 16N-27HZ	Hudson 13N-35HZ	Nichols 26C-5HZ	Palayo 35N-W11HZ	Raymond 38N-29HZ	Strear V 06-63HN	Wardell 2N-29HZ	
Brown 14N-26HZ	Camp 3N-25HZ	Cook 37N-W3HZ	Gobbler 1N-23HZ	Hudson 14N-35HZ	Nichols 26N-5HZ	Paradize 4-15HZ	Raymond Federal 16C-30HZ	Strear V03-73HN	Wardell 3C-20HZ	
Brown 1N-35HZ	Camp 3N-31HZ	Dechant 13N-11HZ	Gobbler 28N-23HZ	Hudson 35C-35HZ	Nichols 27N-5HZ	Powers 1C-23HZ	REI 31-5HZ	Thomason 13C-15HZ	Wardell 3N-20HZ	
Brown 27N-35HZ	Camp 4N-25HZ	Dechant 14C-1HZ	Gobbler 29N-23HZ	Hudson 35N-35HZ	Nichols 28C-5HZ	Powers 28N-27HZ	Rice 1N-33HZ	Thomason 13N-15HZ	Wardell 4G-7HZ	
Brown 28C-35HZ	Camp 4N-31HZ	Dechant 35N-E1HZ	Gobbler 2C-23HZ	Hudson 36C-35HZ	Nichols 28N-5HZ	Powers 29C-27HZ	Rice 27N-33HZ	Thomason 14N-15HZ	Wardell 4N-29HZ	
Brown 28N-35HZ	Camp 9N-30HZ	Dechant 35N-W1HZ	Gobbler 2N-23HZ	Hudson 36N-35HZ	Nichols 29N-5HZ	Powers 29N-27HZ	Rice 28N-33HZ	Thomason 26N-9HZXX	Weichel 13N-11HZ	
Brown 2N-35HZ	Camp Federal 30C-31HZ	Dechant 36N-W1HZ	Gobbler 36N-E22HZ	Jelts 15C-19HZ	Nichols 2N-5HZ	Powers 30C-27HZ	Rice 2C-33HZ	Thomason 27N-E9HZ	Weichel 14N-11HZ	
Brown 35N-26HZ	Camp Federal 30N-31HZ	Dolph 15C-1HZ	Gobbler 36N-W22HZ	Jelts 16N-19HZ	Nichols 30C-5HZ	Powers 30N-27HZ	Rice 2N-33HZ	Thomason 27N-W9HZ	Weichel 1N-11HZ	

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LIST OF THE SIX "OFFENDER WELLS"

WELL NAME	PAD	ROUTE	SITE SERVICE	TAG
CAMP 10N-30HZ	CAMP 37-31HZ	NH5	02	WTTNBERG.UIS02:CAMP_R_10N_30_M1_FI_G_AV
CANNON 13N-11HZ	CANNON 4-11HZ	E34	06	WTTNBERG.UIS06:CANN_S_13N_11HZ_M1_FI_G_AV
COLFER 13C-34HZ	COLFER 4-34HZ	EH1	04	WTTNBERG.UIS04:ALLEN_D_13C_M1_FI_G_AV
GOBBLER 1N-23HZ	GOBBLER 37-23HZ	EZ4	05	WTTNBERG.UIS05:GOBB_A_1N_23_M1_FI_G_AV
WARDELL 26N-29HZ	WARDELL 37-29HZ	EZ1	08	WTTNBERG.UIS08:WARDELL_R_26N_29_M1_FI_G_AV
BROWN 1N-35HZ	BROWN 16-35HZ	NH2	04	WTTNBERG.UIS04:DEJANE_A_29C_36_M1_FI_G_AV